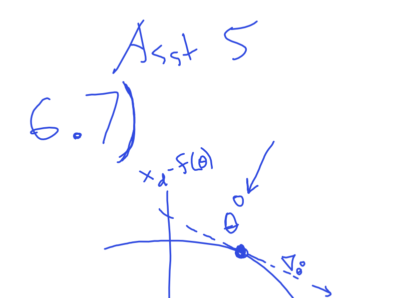
Asst 5

6.7)



the newton-raphson root-finding method causes the algorithm to skip over the first basin of attraction and instead converge on the further basin of attraction. This is because $x_d - f(\theta)$ is a non-convex function

6.8)

$$g(x,y) = \begin{bmatrix} x^2 - 4 \\ y^2 - 9 \end{bmatrix}$$

$\bar{x}^0 = (1, 1)$

we want $g(x,y) = \vec{0}$

$$\nabla g(x,y) = \begin{bmatrix} \nabla_x g_1 & \nabla_y g_1 \\ \nabla_x g_2 & \nabla_y g_2 \end{bmatrix} = \underbrace{\begin{bmatrix} \nabla_x \\ \nabla_y \end{bmatrix}}_{2\times1} \underbrace{\begin{bmatrix} g_1 \\ g_2 \end{bmatrix}}_{2\times1}$$

$$\nabla g = \begin{bmatrix} 2x & 0 \\ 0 & 2y \end{bmatrix} \to \nabla g^{-1} = \begin{bmatrix} \frac{1}{2x} & 0 \\ 0 & \frac{1}{2y} \end{bmatrix}$$

$$\Delta\theta^0 = \underbrace{J^{-1}(\theta^0)}_{\nabla g^{-1}} \underbrace{(x_d - f(\theta^0))}_{g(\theta^0)}$$

$$= \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix} \begin{bmatrix} -3 \\ -8 \end{bmatrix} = \begin{bmatrix} -1.5 \\ -4 \end{bmatrix}$$

$$\theta^1 = \theta^0 - \Delta\theta^0 = \begin{bmatrix} 2.5 \\ 5 \end{bmatrix}$$

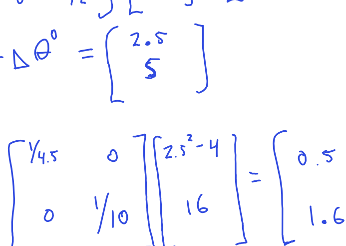
$$\Delta\theta^1 = \begin{bmatrix} 1/4.5 & 0 \\ 0 & 1/10 \end{bmatrix} \begin{bmatrix} 2.5^2 - 4 \\ 16 \end{bmatrix} = \begin{bmatrix} 0.5 \\ 1.6 \end{bmatrix}$$

$$\theta^2 = \theta^1 - \Delta\theta^1 = \begin{bmatrix} 2 \\ 3.4 \end{bmatrix}$$

all ~~rts~~ roots: (2, 3), (−2, 3), (2, −3), (−2, −3)

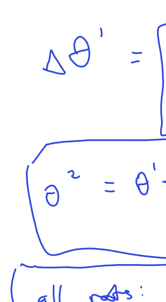
4 roots

8.2)



$$I_b = -\sum_i m_i [r_i]^2 \quad \in \mathbb{R}^{3\times3}$$

Cylinder:



$$I_{xx} = I_{yy} = \frac{m}{12}(3r^2 + h^2)$$
$$I_{zz} = \frac{mr^2}{2}$$

sphere: $I_{xx} = I_{yy} = I_{zz} = \frac{m}{5}(2r^2)$

eq 8.28: $I_d = I_{cm} + md^2$

eq 8.32 $G_b = \begin{bmatrix} I_b & 0 \\ 0 & mI_{d''} \end{bmatrix}$

$$I_{b_{cyl}} = \begin{bmatrix} 0.00647 & 0 & 0 \\ 0 & 0.00647 & 0 \\ 0 & 0 & 0.003015 \end{bmatrix}$$

$m_c = 7500 h \pi r^2$
$= 7500(0.2)\pi(0.02)^2$

$I_{xx} = \frac{m}{12}(3(.02)^2 + 0.2^2)$
$= 0.00647$
$I_{zz} = 0.003015$

$$I_{b_{sph}} = \begin{bmatrix} 0.1256 & 0 & 0 \\ 0 & 0.1256 & 0 \\ 0 & 0 & 0.1256 \end{bmatrix}$$

$m = 4/3\,\pi(0.1)^3 \cdot \rho$
$I_{xx} = 0.256$

two spheres

$$I_{all} = I_{cyl} + 2(I_s + m_{sph} d_{sph}^2)$$

$d_{sph} = (0.0, 0.0, 0.2)$

no change to I_{xx}, I_{yy}

$$I_b = \begin{bmatrix} 2.52 & 0 & 0 \\ 0 & 2.52 & 0 \\ 0 & 0 & 2.76 \end{bmatrix}$$

$$G_b = \begin{bmatrix} I_b & 0 \\ 0 & mI_{d''} \end{bmatrix} = \begin{bmatrix} 2.52 & & & & & \\ & 2.52 & & & 0 & \\ & & 27.6 & & & \\ & 0 & & 64.7 & & \\ & & & & 64.7 & \\ & & & & & 64.7 \end{bmatrix}$$

$m_{all} = m_{cyl} + 2 \cdot m_{sph}$
$= 64.7$

4) $\mathcal{L} = \mathcal{L}^1 + \mathcal{L}^2 + \ldots$

$$\mathcal{L}^1 = m_2 L_1 L_2 \dot\theta_1^2 \cos\theta_2$$

τ_1, τ_2 from $\mathcal{L}^1$? $\quad \tau_i = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot\theta_i} - \frac{\partial \mathcal{L}}{\partial \theta_i}$

eq 8.8

① $\frac{\partial \mathcal{L}}{\partial \dot\theta_1} = 2m_2 L_1 L_2 \dot\theta_1 \cos\theta_2$

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot\theta_1} = 2m_2 L_1 L_2(\ddot\theta_1 \cos\theta_2 - \dot\theta_1 \sin(\theta_2) \cdot \dot\theta_2)$$

$\frac{\partial \mathcal{L}}{\partial \theta_1} = 0$

$M(\theta)\ddot\theta$ term

vel product term

$$\Rightarrow \boxed{\tau_1 = 2m_2 L_1 L_2(\ddot\theta_1 \cos\theta_2 - \dot\theta_1 \sin(\theta_2)\dot\theta_2)}$$

② $\frac{\partial \mathcal{L}}{\partial \dot\theta_2} = 0 \quad \frac{\partial \mathcal{L}}{\partial \theta_2} = -m_2 L_1 L_2 \dot\theta_1^2 \sin\theta_2$

$$\to \boxed{\tau_2 = -m_2 L_1 L_2 \dot\theta_1^2 \sin\theta_2}$$

KE term aka vel product term

no potential terms

5) Forward eqs (8.50 - 8.52)

(1) $T_{i,i-1} = e^{-[A_i]\theta_i} M_{i,i-1}$

(2) $\mathcal{V}_i = Ad_{T_{i,i-1}}(\mathcal{V}_{i-1}) + A_i \dot\theta_i$

(3) $\dot{\mathcal{V}}_i = Ad_{T_{i,i-1}}(\dot{\mathcal{V}}_{i-1}) + ad_{\mathcal{V}_i}(A_i)\dot\theta_i + A_i\ddot\theta_i$

Backward eqs (8.53 - 8.54)

(4) $\mathcal{F}_i = Ad^T_{T_{i+1,i}}(\mathcal{F}_{i+1}) + G_i\dot{\mathcal{V}}_i - ad^T_{\mathcal{V}_i}(G_i\mathcal{V}_i)$

(5) $\tau_i = \mathcal{F}_i^T A_i$

(1) get the kinematic transformation from the current link to the previous link

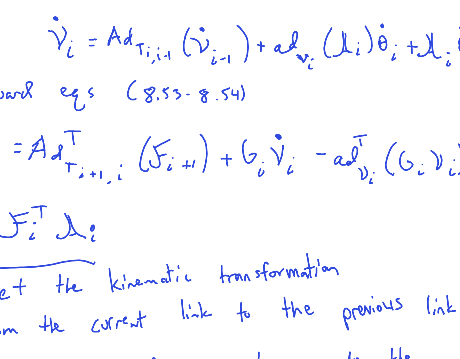
(2) use forward dynamics to compute the twist of each link due to (1) the previous links twist and (2) the current link's joint rotation

(3) compute each link's accelerations, combining
(1) previous link's accelerations
(2) velocity product term (aka coriolis & centripetal)
(3) this link's joint acceleration

(4) compute the wrench at link i due to
(1) upstream wrenches transformed into this link's frame
(2) this link's wrench
(3) velocity product term (aka coriolis & centripetal)

(5) compute the joint torque given the joint wrench & twist.

9.1)



$f(0) = (0,0)$
$f(1) = (0,0)$

let $f(0.25) = (2,1)$
$f(0.5) = (4,0)$
$f(0.75) = (2,-1)$

ellipse eqs: $x = a\cos t + b$
$y = c\sin t + d$

@ $t = 0 = Ts$ $\quad x = a+b$, $y = d$ $= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow d = 0$

@ $t = \pi/2 = Ts$ $\quad \bar{x} = \begin{bmatrix} b \\ c \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \Rightarrow b = 2$, $c = 1$

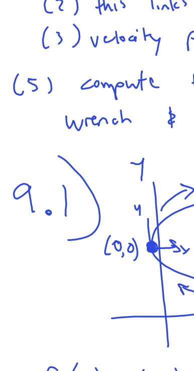
@ $t = \pi = Ts$ $\quad \bar{x} = \begin{bmatrix} -a+2 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \end{bmatrix} \Rightarrow a = -2$

@ $t = 2\pi = Ts$, $s = 1$ $\quad \bar{x} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ ✓

$\to T = 2\pi$

$$\boxed{f(s) = \begin{bmatrix} -2\cos(2\pi s) + 2 \\ 1\sin(2\pi s) \end{bmatrix}}$$

9.3) $X(0) = X_{start} \in SE(3)$
$X(1) = X_{end} \in SE(3)$



$s = s(t)$

$\mathcal{V}$? $\dot{\mathcal{V}}$? given $\dot s$, $\ddot s$

$X_1 = X_0 e^{[S]}$

$\hookrightarrow X_0^{-1} X_1 = e^{[s]}$

$[s] = \log(X_0^{-1} X_1) = $ constant

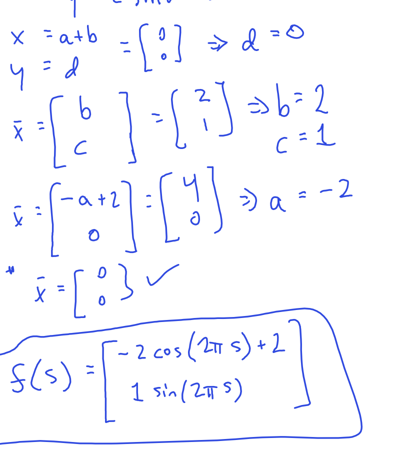
from eq 9.6

now, $X(s) = X_{start}\exp([s]s)$

$$\boxed{\mathcal{V}(s) = \dot X(s) = X_{start}\exp([s]s) \cdot \underbrace{\log(X_0^{-1}X_1)}_{=[s]} \dot s}$$

$$\boxed{\dot{\mathcal{V}}(s) = \ddot X(s) = X_{start}(\exp([s]s) \cdot [s]\dot s) \cdot [s]\dot s + X_{start}\exp([s]s) \cdot [s]\ddot s}$$

9.21)

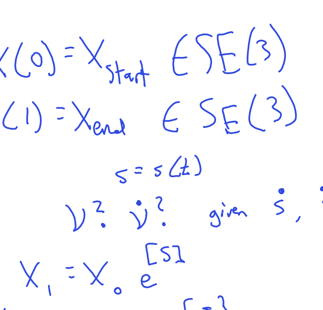


A is not possible because s isn't monotonically increasing

B is not possible because at $\dot s = 0$ there shouldn't be any change in S.

b & c are not possible because at $\dot s = 0$ s can't change

9.22)



always either bang-bang control or tangent to vel limit curve ⇒ time optimal

(a) the test curve uses max deceleration, as does F but starts at an earlier s-value, so it will never intersect F. The only other possibilities are then intersecting $\dot s = 0$ or the vel curve

(b) touching the curve tangentially means it's going as fast as possible without violating the limits.

(c) to be time optimal you always want to either fully accel, fully decell, or be tangent to the vel limit curve.

11.6) 2nd order

$$m\ddot x + b\dot x + kx = f$$

$m = 4$
$b = 2$
$k = 0.1$

$$\hookrightarrow \ddot x + \frac{b}{m}\dot x + \frac{k}{m}x = \frac{f}{m}$$

$\frac{b}{m} = 2\zeta\omega_n \quad \zeta = b/(2\sqrt{km})$

$\frac{k}{m} = \omega_n^2 \quad \omega_n = \sqrt{\frac{k}{m}} = 0.158$

roots $s^{\pm} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$

stable if $\zeta\omega_n > 0$ & $\omega_n^2 > 0$ ✓

a) $\zeta = 2/2\sqrt{0.1 \cdot 4} = \boxed{1.58 > 1 \Rightarrow \text{overdamped}}$

$s_{1,2} = -.056, -0.44$, $t_{1,2} = \frac{-1}{s_{1,2}}$

$t_{1,2} = 17.7, 2.25$

2% s.t. $\approx 4t = \boxed{71\text{ s}}$

(b) $f = k_p x_e \to \ddot x + \frac{b}{m}\dot x + \frac{\overbrace{(k+k_p)}^{k'}}{m}x = f$

if $\zeta = 1 = \frac{b}{2\sqrt{k'm}} \to k' =$

$\sqrt{k'm} = \frac{b}{2}$, $k'm = \frac{b^2}{4}$, $k' = \frac{b^2}{4m} = 4/16 = 1/4 = k + k_p$

$\boxed{\Rightarrow k_p = 0.15}$

(c) $f = k_d\dot x \to b' = b + k_d$

$\to \zeta = 1 = \frac{b'}{2\sqrt{km}} \to b' = 2\sqrt{km} = 1.26$

$\to \boxed{k_d = -0.74}$

(d) PD, $b' = b + k_d$, $k' = k + k_p$

$\zeta = 1 = \frac{b'}{2\sqrt{k'm}} \to b' = 2\sqrt{k'm}$

2% $= 0.01\text{ s} = 3.91t$, $t = 1/\omega_n = 1/\sqrt{\frac{k'}{m}}$

$\to 0.01 = \frac{3.91}{\sqrt{\frac{k'}{m}}} \to \sqrt{\frac{k'}{m}} = \frac{3.91}{0.01}$, $\frac{k'}{m} = 391^2$

$k' = 391^2 \cdot m \to k' = 611{,}524$

$\boxed{\to k_p = 611\,523.9}$

$b' = 2\sqrt{k'm} = 3128 \to \boxed{k_d = 3126}$

(e) $x_d = 1$ $\quad$ laplace: $(ms^2 + bs + k)X(s) = F(s)$

~~$e_{ss} = \frac{1}{\ } = 1.64 \times 10^{-6}$~~ ~~$\dot x_e = 0$~~

~~$\hookrightarrow f_{ss} = k_p e_{ss} = 0.999998$~~

assume $\dot x = 0$, $\ddot x = 0$

@ $t = \infty$, $\dot x = \ddot x = 0 \Rightarrow$

$kx = f = k_p x_e = k_p x = k_p(1-x)$

$\Rightarrow (k_p + k)x = k_p$

$f_{inf} = k_p \cdot e_{ss}$ $\quad \Rightarrow x_\infty = \frac{k_p}{k_p + k} \Rightarrow e_{ss} = 1 - x_\infty$

$$\boxed{\Rightarrow e_{ss} = 1.635e^{-7},\ f_{inf} = 0.09999998}$$

(f) $m\ddot x + b\dot x + kx = k_p x_e + k_d\dot x_e + \underset{\text{maybe}}{k_i}\int x_e\,dt$

let $\dot x_d = \ddot x_d = 0$, $x_d \neq 0$

$x_e = x_d - x \to x = x_d - x_e$

$\dot x_e = -\dot x$ $\quad \hookrightarrow -m\ddot x_e - b\dot x_e + k(x_d - x_e)$

$\ddot x_e = -\ddot x$ $\quad = k_p x_e + k_d\dot x_e + k_i\int x_e\,dt$

$$\hookrightarrow \boxed{kx_d = (k_p + k)x_e + (k_d + b)\dot x_e + m\ddot x_e + k_I\int x_e\,dt}$$

$\frac{d}{dt} \Rightarrow (k_p + k)\dot x_e + (k_d + b)\ddot x_e + m\dddot x_e + k_I x_e = 0$ $\quad$ 3rd order error dynamics

all roots must have negative real component for stability

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$$\boxed{k_p + k > 0,\quad k_d + b > 0,\quad k_i > 0,\quad \frac{(b+k_d)(k+k_p)}{m} > k_i}$$

$$\boxed{\text{when } t \to \infty,\ \dot x_e = \ddot x_e = \dddot x_e = 0 \to 0 = k_i x_e \Rightarrow x_e = 0\ ✓}$$